**1** **(a)** 17.5 (or $17\frac{7}{15}$ or 17.46...) **[B1]**

**(b)** 17 **[B1]**

**(c)** Upper quartile = 21 and Lower quartile = 15 **[M1]**

6 **[A1]**

*[Total Marks: 4]*

**2** **(a)** **(i)** 3.0476... **[M1]**

3.05 **[A1]**

**(ii)** 3 **[B1]**

**(iii)** Upper quartile = 4 and Lower quartile = 2 identified **[M1]**

2 **[A1]**

*[Total Marks: 5]*

**3** **(a)** 16 **[B2]**

If B0, M1 for $\sum x = 240$ seen

**(b)** 15 **[B1]**

**(c)** 6 **[B2]**

If B0, M1 for $Q_1 = 13$ and $Q_3 = 19$ seen

*[Total Marks: 5]*

**4** **(a)** $20 < t \leq 30$ **[B1]**

**(b)** Use of correct midpoints: $5, 15, 25, 35, 50$ **[M1]**

$\Sigma fx = 14(5) + 28(15) + 35(25) + 18(35) + 5(50)$ or 2245 seen **[M1]**

22.45 **[A1]**

**(c)** The exact times are not known so the midpoints of the intervals are used. **[B1]**

*[Total Marks: 5]*

**5** **(a)** Use of correct mid-interval values $(5, 15, 25, 35, 50)$ **[M1]**

$\Sigma ft = 5(12) + 15(25) + 25(28) + 35(11) + 50(4)$ or 1720 **[M1]**

21.5 **[A1]**

**(b)** The exact times are not known / the data is grouped so mid-interval values are used. **[B1]**

**(c)** The mean will decrease because 12 is less than the current mean (21.5). **[B1]**

*[Total Marks: 5]*

**6** **(a)** 21.25 **[B2]**

If B0, M1 for $\sum fx = 1700$ or midpoints $5, 15, 25, 35, 45$ seen

**(b)** The exact times are not known / the midpoints of each class are used as an estimate. **[B1]**

**(c)** $120 \times 24.5 = 2940$ **[M1]**

$(1700 + 2940) \div 200$ **[M1]**

23.2 **[A1]**

*[Total Marks: 6]*

**7** **(a)** $3 + x + y + 6 + 4 + 1 = 35$ leading to $x + y = 21$ **[M1]**

Expression for total goals: $0(3) + 1(x) + 2(y) + 3(6) + 4(4) + 5(1)$ or $x + 2y + 39$ **[M1]**

$\dfrac{x + 2y + 39}{35} = 2.2$ leading to $x + 2y = 38$ **[M1]**

$x = 4$ and $y = 17$ **[A1]**

**(b)** 2 **[B1]**

*[Total Marks: 5]*

**8** **(a)** $\dfrac{0 \times 6 + 1 \times x + 2 \times 8 + 3 \times 4 + 4 \times 2}{6 + x + 8 + 4 + 2} = 1.5$ **[M1]**

$\dfrac{x + 36}{x + 20} = 1.5$ **[M1]**

$x + 36 = 1.5x + 30 \implies 0.5x = 6 \implies x = 12$ **[A1]**

**(b)** 1 **[B1]**

**(c)** 1 **[B2]**

If B0, M1 for $Q_1 = 1$ and $Q_3 = 2$ seen

*[Total Marks: 6]*

**9** **(a)** Total frequency $= 10 + k + 2k + 3 + 7 + 5 = 3k + 25$ **[M1]**

Total score $= 10(1) + k(2) + (2k + 3)(3) + 7(4) + 5(5) = 8k + 72$ **[M1]**

$\dfrac{8k + 72}{3k + 25} = 2.8$ **[M1]**

$8k + 72 = 8.4k + 70 \implies 0.4k = 2 \implies k = 5$ **[A1]**

**(b)** **(i)** 3 **[B1]**

**(ii)** 1.5 **[B1]**

**(iii)** 4 **[B1]**

*[Total Marks: 7]*